

Program : Diploma in Fire Technology and Safety	
Course Code: 3464	Course Title: Electrical Safety
Semester: 3	Credits: 4
Course Category: Program Core	
Periods per week: 4 (L:3 T:1 P:0)	Periods per semester: 45

Course Objectives:

- To understand the principles and importance of electrical safety in various environments.
- To identify the causes and prevention of electrical shocks.
- To ensure safety measures during electrical installation, testing, and maintenance.
- To learn about electrical safety in hazardous areas and fire safety procedures

Course Pre-requisites:

Topic	Course Code	Course Name	Semester
Principles of Safety Management		Principles of Safety Management	2

Course Outcomes

On completion of the course, the student will be able to:

CO _n	Description	Duration (Hours)	Cognitive Level
CO1	Identify electrical hazards and apply safety measures to prevent electric shocks.	10	Understanding
CO2	Apply safety precautions in residential, commercial, and agricultural installations.	11	Applying
CO3	Understand the procedures for electrical safety during installation, testing, and maintenance.	10	Applying
CO4	Understand electrical safety measures in hazardous areas and fire extinguishing techniques.	12	Understanding

	Series Test	2	
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CO – PO Mapping:

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3	2	3	2			
CO2	3	3	2	3			
CO3	3	2	3	2			
CO4	3	3	3	3			

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline

Module Outcomes	Description	Duration (Hours)	Cognitive level
CO1	Understand electrical hazards and the methods to prevent electric shocks.		
M1.01	Terms and definitions in electrical safety and hazards.	2	Understanding
M1.02	Types of electrical shocks and their effects on the human body.	3	Understanding
M1.03	Prevention of electrical shocks and safety measures.	5	Applying
Contents: Introduction to Electrical Safety, Shocks and Their Prevention Electrical safety objectives and principles. Types of shocks: Primary and secondary electrical shocks. Causes, severity, and effects of electrical shocks. Prevention measures for contact shocks, flash shocks, burns. Safety precautions in residential and commercial buildings.			
CO2	Apply electrical safety precautions for various types of installations.		
M2.01	Safety measures for residential electrical installations	5	Understanding

M2.02	Electrical safety in commercial and agricultural installations.	6	Understanding
	Series Test I	1	
Contents: Electrical Safety in Residential, Commercial and Agricultural Installations Wiring and fitting safety in residential buildings. Safety in the use of domestic electrical appliances. Electrical safety in multi-storied buildings and temporary installations. Agricultural pump installation safety. Do's and Don'ts for safety in residential and commercial settings.			
CO3	Implement electrical safety measures during installation, testing, and maintenance.		
M3.01	Safe installation and testing procedures for electrical systems.	6	Applying
M3.02	Safety documentation, PPE, and safety clearance during operations.	4	Applying
Contents: Electrical Safety during Installation, Testing, and Commissioning Safe sequence of installation and testing. Risk management in electrical plants and equipment. Safety documentation and field safety quality. Use of personal protective equipment (PPE). Safety precautions for operators during electrical operations.			
CO4	Understand safety measures in hazardous zones and fire safety techniques.		
M 4.01	Electrical safety in hazardous zones (Class 0, 1, 2).	5	Understanding
M 4.02	Fire safety: types of fires, extinguishing methods, and fire safety equipment.	7	Understanding
	Series Test II	1	

Contents: Electrical Safety in Hazardous Areas and Fire Extinguishers

Classification of hazardous zones and electrical equipment for hazardous locations.
Specifications of equipment enclosures for hazardous gases and vapors.
Fire initiation, types, and extinguishing techniques.
Fire extinguishers: CO₂, foam, and halogen gas schemes.
Fire detection and alarm systems.

Text /Reference:

T/R	Book Title/Author
T1	Electrical Safety Engineering - J. C. Das (McGraw-Hill Education).
T2	Electrical Safety Handbook - H. P. S. Kalsi (McGraw-Hill Education).
R1	Electrical Safety: A Guide to Safe Installation and Operation - Dennis K. McBride.
R2	Fundamentals of Electrical Safety - A. K. Gupta (Khanna Publishers).